MIL-I-22960A(SH)
8 January 1987
SUPERSEDING
MIL-I-22960(SHIPS)
23 November 1961
(See 6.6)

MILITARY SPECIFICATION

RADAR INDICATORS FOR NAVAL SURFACE SHIPS, SUBMARINES AND SHORE INSTALLATIONS, GENERAL SPECIFICATION FOR

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and definitions applicable to the construction of radar indicators with cathode ray tube (CRT) displays for Naval surface ships, submarines, and shore installations. These radar indicators are not an integral part of the radar transmitter-receiver system or a part of a weapons system.

1.2 Configuration. Radar indicators covered by this specification shall be for displaying radar echo signals from navigation, surface and air surveillance radar sets in a plan position indicator (PPI)/range height indicator (RHI) presentation (see 6.3.4 and 6.3.5), by a scanning method as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-E-1 - Electron Tubes, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL-I-983	- Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.
MIL-E-16400	- Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-D-87157	- Display, Diode, Light Emitting, Solid State, General Specification for.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-242	- Electronic Equipment Parts, Selected Standards.
MIL-STD-242/8	- Electronic Equipment Parts, Selected Standards, Connectors.
MIL-STD-415	- Test Provisions For Electronic Systems and Associated Equipment, Design Criteria for.
MIL-STD-454	- Standard General Requirements For Electronic Equipment.
MIL-STD-751	- Radar Outputs, Naval Ship and Shore.
MIL-STD-1311	- Test Methods for Electron Tubes.
MIL-STD-1348	- Knobs, Control, Selection of.
DOD-STD-1399, Section 300	- Interface Standard For Shipboard Systems, Electric Power, Alternating Current. (Metric)
MIL-STD-1472	- Human Engineering Design Criteria For Military Systems, Equipment and Facilities.
MIL-STD-1553	- Aircraft Internal Time Division Command/Response Multiplex Data Bus.

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

TE-000-AB-GTP-010	- Electronic Parts Application Manual.
NAVSEAINST 3900.2A	- Reliability and Maintainability of the Naval Sea Systems Command for Design, Development and Acquisition.

NAVAL MATERIAL COMMAND (NAVMAT)

0518-LP-098-2000	- Navy Power Supply Reliability Design and Manufacturing Guidelines.

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(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. The radar indicator (see 6.3.6) shall be a two dimensional, intensity-modulated indicator to display the radar outputs from Navy surface ship, submarine or shore installation search radar sets. The radar indicator shall be stand-alone and shall interface with external processors and controls (see 6.2). Where specified, the recommended interface is MIL-STD-1553. The radar indicator shall meet the requirements of MIL-E-16400.

3.2 Display requirements.

3.2.1 Display element. The display element for radar indicators shall be a CRT.

3.2.2 CRT specification. CRTs shall be the militarized version and shall be constructed and tested in accordance with MIL-E-1. The CRT shall have, as a minimum, the following characteristics:

(a) The tube envelope size shall be as specified (see 6.2). Recommended PPI area shall be 10 inches in diameter for detection and tracking.
(b) The phosphor screen shall have a usable area as specified (6.2).
(c) The screen shall be a Joint Electron Devices Engineering Council registered aluminized phosphor.
(d) The CRT shall operate continuously within its high voltage rating on the anode, without arc-over.
(e) The CRT shall be high voltage electrostatically focused.
(f) The electron beam shall be deflected by magnetic deflection.
(g) The line width shall be not less than the human eye's ability to resolve and shall be measured at the 50 percent point of intensity in both the horizontal and vertical direction at the maximum writing speed of operation (see 6.2).
(h) High voltage dynamic focus shall be used for uniform focusing in the specified equipment.
(i) A glass safety panel shall be bonded to the CRT faceplate for implosion protection.
(j) The outside surface of the bonded safety panel shall have a high efficiency anti-reflection coating, which shall also be an electromagnetic shield when grounded in the indicator.
(k) The combination of the safety panel, bonding, CRT glass faceplate and anti-reflection coating shall maximize brightness and contrast for operating in a daylight environment without use of a viewing hood.

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(l) Militarized shall mean that the CRT shall operate in the indicator during and after shock, vibration, and environmental testing.
(m) Room temperature vulcanizing (RTV) and silastic compounds used to encapsulate the anode connection and the tube base leads shall be primed and cured for complete adherence to components and leads.
(n) The CRT assembly shall be chosen so that the assembly can be changed in the indicator without disassembly of any connector or unsoldering of connectors.

3.2.3 Video system response. The CRT size, CRT line width, video amplifier and signal source sensor shall provide a total modulation transfer function (MTF) (see 6.3.3) system spurious response of 25 percent when the system is displaying radar returns consisting of pulses as short as 50 nanoseconds duration and observed by the operator at a distance of 20 inches.

3.2.4 CRT phosphor. The phosphor for the CRT shall be selected for color, efficiency, life, and persistence. Phosphors shall be aluminized. Persistence shall be compatible with the scanning system. Resolution requirements shall be within the capability specified for the phosphor (see 6.2).

3.2.5 CRT X-ray threshold. The threshold generation of X-rays in CRTs shall be considered to be from an electron beam accelerated at or greater than 16 kilovolts (kV). When an accelerating voltage exceeding 16 kV is specified, X-ray suppression shall be in accordance with requirement 1 of MIL-STD-454.

3.2.6 Spot burn protection. CRT control circuits shall have a time delay circuit that prevents the premature application of high voltage to the anode and focus electrodes of the CRT. The time delay circuit shall be activated whenever primary alternating current (ac) power to the indicator is initially turned on, and shall function and recycle whenever primary power to the indicator is interrupted for longer than the predetermined warm-up period. The time delay circuit shall not allow anode and focus voltage to be applied to the CRT until the operational circuits of the indicator have stabilized during a predetermined warm-up time. The time delay circuit shall be interlocked with a deflection current sensing circuit. This circuit shall prevent the anode and focus voltage from being applied to the CRT when the power off time is longer than the warm-up time and X and Y deflection currents are not flowing in the deflection yoke. The deflection current sensing circuit shall cause a bias voltage to be applied to the CRT that shall cut off the electron beam at any time in the absence of deflection current. When the deflection currents are again present in the deflection yoke, anode and focus voltage shall be applied to the CRT.

3.2.7 High voltage power supply protection. The high voltage power supply for the CRT anode and focus electrodes shall be protected from over current damage by a current limiting circuit within the high voltage power supply (not a series limiting resistor). If a separate high voltage power supply is used for CRT dynamic focus, it shall be protected with a current limiting circuit.

3.2.8 High voltage power supply regulation. The high voltage power supply load regulation shall not exceed 0.05 percent at the rated CRT voltage and current.

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3.2.9 Magnetic shielding. Magnetic shielding for the CRT shall be provided to prevent the effect, on the CRT display, of external magnetic fields in accordance with requirement 60 of MIL-STD-454.

3.2.10 Critical fusion frequency. The radar display shall be refreshed at a frequency higher than the frequency at which the critical fusion frequency (CFF) or flicker (see 6.3.2) occurs. The CFF shall be calculated using the following formula:

$$f_c = 12.57 \log_{10} A + 34$$

where:

f_c = critical fusion frequency

A = brightness of photopic highlights in display, footlamberts

This formula is applicable for CRT displays with medium to fast phosphor decay. Refresh rate shall be greater than the CFF caluculated from the formula and adjusted upwards to a rate which is technically compatible with the requirements of the Navy and illumination of the operating environment.

3.3 Deflection circuitry.

3.3.1 Deflection. The electron beam of the CRT shall be magnetically deflected. A single fixed position, ruggedized, high precision, encapsulated, stator type yoke shall be used. The circuitry for the yoke driver transistors shall be so configured that matching yoke driver transistors are not required. The deflection yoke windings shall be critically dampened to the lowest deflection frequency to prevent ringing in the display. Dampening shall be accomplished at the yoke. The wiring between the deflection amplifier and the yoke shall be as short as possible and completely shielded from extraneous fields to eliminate ringing and distortion in the trace. If the yoke is not encapsulated in the CRT assembly, the yoke shall be removable from the CRT without disassembly of wiring and shall be connected to the deflection amplifier with a connector.

3.3.2 Luminance shades. The display system shall display to the operator brightness variations of the display equivalent to a minimum of eight luminous shades (see 6.3.7).

3.3.3 Contrast ratio. The contrast ratio of the display shall be at a minimum, equal to $D = C^m$, where:

D = contrast ratio
C = contrast change detected by human eye, $(2)^{1/2}$
m = number of contrast changes (luminous shades, see 3.3.2)

Thus, at a minimum, a display with eight luminous shades shall have a contrast ratio of 16 to one for a daylight environment.

3.3.4 Resolution requirement. MTF shall be used as the criteria for the resolution requirement for radar indicators. The CRT display shall be configured and specified as having an MTF of 50 percent.

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3.3.5 Video amplifier. The amplitude of the video signal from the radar set to the video input of the indicator shall be in accordance with MIL-STD-751. The CRT video amplifier shall have uniform frequency response within 0.5 decibel (dB) for a frequency range from direct current (dc) to the inverse of twice the rise time (t_r) of the resolution points generated by the deflection system. The frequency response shall be measured with the CRT video amplifier connected to and driving the control element of the CRT display. The gain of the CRT video amplifier shall be sufficient to drive the CRT to zero bias with a 2 volt peak amplitude video input signal and maintain uniform frequency response as specified herein for all gain settings. The frequency response shall not have a sharp cut-off characteristic at the limit of the uniform frequency response, but shall have a remote roll-off response. Restoration to dc shall be provided to reference the control electrode of the CRT to a fixed dc bias voltage to maintain uniform black level.

3.3.6 Signal-to-noise ratio. The signal-to-noise ratio at the output of the video amplifier, when measured at the modulation electrode of the CRT, shall be greater than 35 dB.

3.3.7 Blanking. Blanking (see 6.3.1) of the video signal shall occur prior to display deflection retrace and continue past the active trace time before unblanking.

3.3.8 Analog video input signals. The radar indicator shall operate with analog video signals that are in accordance with MIL-STD-751. When measured at the terminated video input connector of the indicator, the video signals shall be of positive polarity with a peak amplitude of 1 to 3 volts.

3.3.8.1 Analog video input connectors. A coaxial cable connector receptacle selected in accordance with MIL-STD-242/8 shall be provided in the radar indicator to accommodate the incoming analog video.

3.3.9 Digital video input. When specified (see 6.2) the radar indicator shall accept digitized video signals. The radar indicator shall have a digital to analog converter (D/A) with a conversion rate to accommodate the frequency response.

3.3.10 Composite video output. When specified (see 6.2) an output receptacle selected in accordance with MIL-STD-242/8 shall be installed on the radar indicator for the transmission of composite video. The output signal shall be in accordance with MIL-STD-751 and shall drive a terminated 75 ohm coaxial cable for operating remote radar indicators.

3.4 Digital data interface.

3.4.1 Radar display and distribution system (RADDS) serial data stream. Radar indicators shall accept, process and operate with the RADDS serial data stream in accordance with MIL-STD-751 (see 6.3.9).

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3.5 Built-in-test. Unless otherwise specified (see 6.2), radar indicators shall provide for monitoring performance, calibration and fault isolation by built-in-test (BIT) under control of the display controller. Monitoring shall be performed continuously on power supplies, plug-in boards and synchro-to-digital converters. The following test shall be performed as a minimum:

(a) End-to-end dynamic system targets.
(b) Test patterns.
(c) Electronic cursor tests.
(d) All readouts.
(e) Offsets.
(f) Controls.
(g) Lamp test.
(h) Test of decay rates.
(i) Test of all data storage (both dynamic and static).
(j) Controllers.

3.6 Plan position indicator.

3.6.1 Plan position indicator (type PPI). The plan position or range azimuth indicator shall be a two dimensional, intensity-modulated radar indicator to display the radar outputs that are in accordance with MIL-STD-751 from Navy ship or shore search radar sets by a scanning method as specified (see 6.2).

3.6.2 Range control. A variable logarithmic range control shall be provided. Range indicators shall be in error no more than 5 percent under all operating conditions.

3.6.3 Range transmission. When specified (see 6.2), provisions for range transmission shall be provided.

3.6.4 Bearing transmission. When specified (see 6.2), provisions for bearing transmission shall be provided.

3.6.5 Gated range trigger output. When specified (see 6.2), an output receptacle selected in accordance with MIL-STD-242/8 shall be installed on the indicator for the transmission of the gated range trigger.

3.6.5.1 Bearing circuits. The bearing circuits shall operate without visual irregular movement of the PPI trace.

3.6.6 Bearing and range indication. Bearing and range indication accuracy shall be as specified (see 6.2).

3.6.7 Bearing circle. A calibrated numbered bearing circle of 360 degrees shall be provided.

3.6.8 True bearing indication. An indication shall be provided to show in what mode the indicator is operating and that the radar is providing true or relative bearing signals.

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3.7 Electrical power.

3.7.1 Primary power. The radar indicator shall operate from electrical power with the characteristics and meet the user equipment constraints of DOD-STD-1399, section 300.

3.7.2 Radar indicator power supplies. The radar indicator shall have its own self-contained high voltage and low voltage power supplies. The high voltage and low voltage power supplies shall conform to the guidelines of 0518-LP-098-2000.

3.7.3 Radar indicator power supply meter. A dc meter and selector switch shall be provided to permit measuring power supply voltages and currents. The dc meter shall permit measurement of power supply voltages to an accuracy of plus or minus 2 percent of the scale meter reading. The meter shall conform to the requirements of MIL-E-16400.

3.7.3.1 Time totalizing meter. A time totalizing meter shall be provided in accordance with MIL-E-16400.

3.7.4 Power supply overload protection. Power supply overload protection shall be in accordance with requirements 1 and 8 of MIL-STD-454 (see 3.3.7).

3.8 Test points. Test points shall be provided in accordance with MIL-STD-415.

3.9 Physical characteristics.

3.9.1 Enclosure. The radar indicator enclosure shall be in accordance with the requirements of MIL-E-16400, with the degree and type of enclosure as specified (see 6.2).

3.9.2 Size. Unless otherwise specified, the size of the radar indicator shall be in accordance with MIL-E-16400 for submarine and surface ships installations.

3.9.3 Accessibility. Accessibility for the radar indicator shall be in accordance with MIL-E-16400.

3.9.4 Internal temperature. The internal temperature within the radar indicator shall not exceed 25 degrees Celsius (°C) above the outside ambient temperature.

3.9.5 Dark adaption. The units shall prevent the escape of undesired light from the radar indicator in accordance with MIL-E-16400.

3.9.6 Air filters. Air filters, if specified for the radar indicator (see 6.2), shall be easily replaceable, shall not be located on the bottom surface of deck mounted units, and shall be in accordance with MIL-E-16400.

3.9.7 Light emitting diode (LED) readouts. When specified (see 6.2), LED readouts used in the radar indicators shall be in accordance with MIL-D-87157.

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3.9.8 Radar selector switch. The radar indicator shall include a radar selector switch which shall control the selection of the radar set from the switchboard. The radar selector switch shall be compatible with radar switchboards used in the Navy. Transient protection of the radar selector switch shall be provided via appropriate circuitry.

3.9.9 Radar video selector switch. The radar indicator shall include a video selector switch which shall control the selection of the radar video signal from the switchboard. The video selector switch shall be compatible with radar switchboards used in the Navy. Transient protection for the video selector switch shall be provided via appropriate circuitry.

3.9.10 Motor driven switches. Motor driven switches shall not be used in the radar indicator.

3.9.11 Stiffstick (joystick). When specified (see 6.2), one or more stiffsticks (joysticks) (see 6.3.8) shall be provided to control the positioning of the electronic cursor and principal designator, and shall maintain enclosure integrity.

3.9.12 Tactile control knobs. Tactile control knobs used on the radar indicator shall be selected in accordance with MIL-STD-1348.

3.9.13 Delay lines. Delay lines shall not be used in the video circuitry of the radar indicator.

3.9.14 Controls. Radar indicator controls shall be in accordance with requirement 28 of MIL-STD-454.

3.9.15 Panel layout. The panel layout for the radar indicator shall be in accordance with requirement 28 of MIL-STD-454.

3.9.16 Lettering. The dials and controls of the radar indicator shall be lettered and numbered in white characters on a black background in accordance with MIL-STD-1472.

3.9.17 Indices. The indices for controls and dials of the radar indicator shall be in accordance with MIL-STD-1472.

3.9.18 Interior controls. The interior switches and controls of the radar indicator shall be provided to facilitate periodic maintenance, alignment and test of the radar indicator and shall be mounted in accordance with requirement 28 of MIL-STD-454. The selection of the switches and controls shall be in accordance with MIL-STD-1472.

3.10 Material. Material used in construction of radar indicators shall be selected in accordance with MIL-E-16400.

3.10.1 Toxic materials. In accordance with MIL-I-983, materials which may produce harmful toxic vapors under conditions encountered in Naval service shall not be used.

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3.10.1.1 Mercury and radioactive material. The use of mercury and radioactive materials shall be in accordance with MIL-E-16400.

3.11 Safety. Safety requirements for the radar indicator, for radiation hazard, electrical hazards, marking and labeling of hazardous parts and materials shall be in accordance with requirement 1 of MIL-STD-454.

3.12 Reliability. The reliability of the radar indicator shall be in accordance with NAVSEAINST 3900.2A.

3.13 Maintainability. The maintainability of the radar indicator shall be in accordance with NAVSEAINST 3900.2A.

3.14 Parts derating. The derating of electronic parts used in the radar indicator shall be in accordance with TE-000-AB-GTP-010.

3.15 Workmanship. The workmanship of the radar indicator shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality assurance provisions. Quality assurance provisions shall be in accordance with MIL-E-16400.

4.3 Sampling. The radar indicator shall be sampled for quality assurance inspection in accordance with MIL-STD-105.

4.4 Tests. The radar indicator shall be tested to validate conformance to the requirements of section 3 in accordance with MIL-E-16400.

4.4.1 CRT. The CRT shall be tested for compliance with MIL-E-1 in accordance with test methods as specified in MIL-STD-1311 (see 3.2.2).

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4.4.1.1 Arc-over. The CRT shall be tested in accordance with MIL-STD-1311, test method 5201, without arc-over (see 3.2.2).

4.4.1.2 Line width. The CRT shall be tested in accordance with MIL-STD-1311, test method 5226, for line width modified for photometer readout (see 3.2.2).

4.4.1.3 Brightness. The CRT shall be tested in accordance with MIL-STD-1311, test method 5221, for brightness and contrast (see 3.2.2).

4.4.1.4 Modulation transfer function. The CRT shall be tested in accordance with MIL-STD-1311, test method 5422, modified for non-storage cathode ray tubes (see 3.2.3).

4.4.1.5 X-ray. The CRT shall be tested in accordance with MIL-STD-1311, test method 5001, for X-radiation measurement (see 3.2.5).

4.4.1.6 Luminance shades. The CRT shall be tested in accordance with MIL-STD-1311, test method 5419, modified for non-storage cathode ray tubes (see 3.3.2).

4.4.2 CRT spot burn protection. The spot burn protection circuits shall be tested to determine that the radar indicator CRT is protected against phosphor burn caused by premature application of focus and accelerating voltages to the CRT before there is deflection current flowing in the deflection yoke requirements as specified in 3.2.6.

4.4.3 High voltage power supply regulation. The high voltage power supply shall be tested by operating the power supply at its rated voltage while it is intermittently loaded and unloaded to its rated current. The variation of the high voltage under these test conditions shall not vary by more than the requirements as specified in 3.2.8.

4.4.4 Signal-to-noise. The signal-to-noise ratio (peak-to-peak signal to RMS noise) of the video amplifier driving the CRT shall be tested for compliance with the signal-to-noise ratio requirements as specified in 3.3.6.

4.5 Inspection of packaging. Sample packs and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation, packing, and marking. Indicators shall be preserved level A, C, or commercial, packed level A, B, C, or commercial (see 6.2) and marked in accordance with MIL-E-17555. Hazardous materials shall conform to the Department of Transportation regulations cited in MIL-E-17555.

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6. NOTES

6.1 Intended use. The radar indicators covered by this specification are intended for use in the configuration and construction of indicators with CRT displays for Naval surface ships, submarines, and shore installations.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Scanning method (see 1.2 and 3.6.1).
(c) Digital data communicator system (see 3.1).
(d) CRT tube envelope (see 3.2.2(a)).
(e) Phosphor screen area (see 3.2.2(b)).
(f) Maximum writing speed of operation (see 3.2.2(g)).
(g) Phosphor resolution requirements (see 3.2.4).
(h) If a digital video input is required (see 3.3.9).
(i) If a video output receptacle is required (see 3.3.10).
(j) If a BIT is not required (see 3.5).
(k) If range transmission is required (see 3.6.3).
(l) If bearing transmission is required (see 3.6.4).
(m) If a gated range trigger output receptacle is required (see 3.6.5).
(n) Bearing and range indication accuracy (see 3.6.6).
(o) Degree and type of enclosure (see 3.9.1).
(p) If air filters are required (see 3.9.6).
(q) If LED readouts are required (see 3.9.7).
(r) If a stiffstick (joystick) is required (see 3.9.11).
(s) Level of preservation and packing required (see 5.1).

6.3 Definitions.

6.3.1 Blanking. Blanking is the process of biasing off the electron beam of a CRT during retrace.

6.3.2 Critical fusion frequency (CFF). CFF is the frequency of intermittent stimulation of the eye at which flicker disappears.

6.3.3 Modulation transfer function (MTF). MTF is the sine-wave frequency amplitude response of sensors and displays.

6.3.4 Plan position indicator (PPI). A PPI is a radar indicator on which targets appear as spots at the same location as they would on a circular map of the area being scanned.

6.3.5 Range height indicator (RHI). A RHI is a radar indicator designed to present height data from a height finding radar system.

6.3.6 Radar indicator. A radar indicator is any display, with its associated equipment, used to present a visual presentation of the echo signals received by a radar set.

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6.3.7 Luminance shades. Luminance shades are the number of different luminance levels that the human eye can distinguish on a display. Eight to ten shades on a military radar is adequate for a high probability of recognition.

6.3.8 Stiffstick (joystick). A stiffstick (joystick) is a two-axis displacement control for XY positioning of an electron beam.

6.3.9 Radar display and distribution system (RADDS). The RADDS serial data stream is a composite data signal consisting of digitized radar information and digitized ship's information.

6.4 Subject term (key word) listing.

Cathode ray tubes
Echo signals
Indicators
Radar

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when materials and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5840-N215)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22960A(SH)	RADAR INDICATORS FOR NAVAL SURFACE SHIPS, SUBMARINES AND SHORE INSTALLATIONS, GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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